IoT-Enabled Cantilever Beam Load Measurement System Using Strain Gauges

Electronics Workshop II - Project Report

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1 Introduction

Load and weight measurement is a fundamental requirement across a wide range of engineering applications, including structural health monitoring, industrial automation, weighing systems, and biomechanical research. Resistive strain gauges remain one of the most widely used transducers for such measurements due to their high sensitivity, small form factor, and well-understood electromechanical behaviour.

The fundamental operating principle of a resistive strain gauge is based on the piezoresistive effect: when a conductive element is mechanically deformed, its electrical resistance changes in proportion to the applied strain. For a metallic foil gauge, this relationship is characterised by the Gauge Factor (GF), defined as:

$$GF = \frac{\Delta R/R}{\varepsilon} \tag{1}$$

where ΔR is the change in resistance, R is the nominal resistance, and ε is the mechanical strain. For standard metallic foil gauges, GF ≈ 2.

Despite widespread use, practical strain gauge systems present significant engineering challenges. The output signal of a Wheatstone bridge driven by a 5 V excitation and loaded with a typical foil gauge is in the millivolt range, necessitating high-gain, low-noise amplification before digitisation. Furthermore, thermal effects, common-mode interference, and bridge imbalance from manufacturing tolerances must be carefully managed.

This work addresses these challenges through a complete bottom-up system design, encompassing: (1) a half-bridge Wheatstone configuration for temperature-compensated differential sensing; (2) a custom discrete instrumentation amplifier for high-gain differential amplification; (3) an active signal conditioning stage for offset correction and low-pass filtering; and (4) an IoT subsystem for wireless data acquisition and live dashboard visualisation.

The remainder of this paper is structured as follows. Section 3 covers the Wheatstone bridge design and analysis. Section 4 describes the instrumentation amplifier. Section 5 presents the signal conditioning stage. Section 6 details the IoT implementation. Section 8 presents experimental results, and Section **??** concludes the paper.

2 System Overview

The complete system signal chain is illustrated in Fig. 1. The strain gauges are bonded to the top and bottom surfaces of an aluminium cantilever beam. When a load is applied at the free end, one gauge experiences tensile strain and the other compressive strain, producing a differential resistance change. This differential signal is extracted by the Wheatstone bridge, amplified by the instrumentation amplifier, conditioned by the summing and filtering stage, digitised by the HX711, and transmitted wirelessly by the ESP32 to a local MQTT broker and Node-RED dashboard.

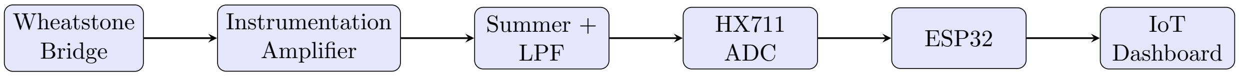

Figure 1: System block diagram showing the complete signal chain from mechanical strain to IoT dashboard.

3 Wheatstone Bridge

3.1 Configuration and Motivation

The Wheatstone bridge is the fundamental transduction interface that converts the resistance change of the strain gauge into a measurable differential voltage. A half-bridge configuration was selected, comprising two active strain gauges (SG1, SG2) and two fixed resistors (R_1, R_2), as shown in Fig. 2.

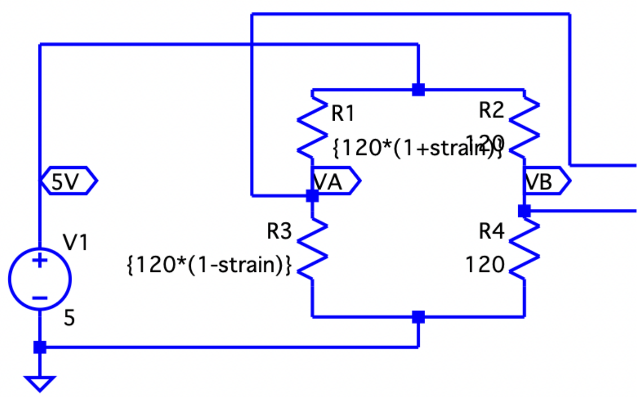

Figure 2: Half-bridge Wheatstone bridge schematic from LTspice. SG1 is bonded to the tensile face and SG2 to the compressive face of the cantilever beam. Fixed resistors $R_1 = R_2 = 100\,\Omega$.

The half-bridge configuration was preferred over a quarter-bridge for two key reasons:

1. **Doubled sensitivity:** With one gauge in tension and the other in compression, both contribute additively to the differential output, doubling the effective bridge output compared to a quarter-bridge arrangement.

2. **Passive temperature compensation:** Since both gauges are bonded to the same beam and experience the same ambient temperature, thermally induced resistance changes are common-mode and cancel differentially, significantly reducing thermal drift in the output signal.

3.2 Bridge Analysis and Output Voltage

The nominal resistance of the strain gauges was measured at $121.8\,\Omega$. Under applied load, the gauges exhibit a resistance change of approximately $\pm 0.5\,\Omega$, corresponding to a strain of:

$$\varepsilon = \frac{\Delta R/R}{GF} = \frac{0.5/121.8}{2} \approx 2050\,\mu\varepsilon \tag{2}$$

The differential output voltage of the half-bridge, with excitation voltage $V_{ex} = 5\,\mathrm{V}$ (from the ESP32 VIN pin), is given by:

$$V_{diff} = \frac{\Delta R}{4R} \cdot V_{ex} \tag{3}$$

For $\Delta R = 0.5\,\Omega$, $R = 120\,\Omega$, $V_{ex} = 5\,\mathrm{V}$:

$$V_{diff} = \frac{0.5}{4 \times 120} \times 5 \approx 5.2\,\mathrm{mV} \tag{4}$$

This millivolt-level output confirms the necessity of high-gain amplification in the subsequent stage.

3.3 LTspice Simulation

The bridge was simulated in LTspice using a parametric resistance sweep to emulate strain variation:

```
.param strain=0
.step param strain -0.001 0.001 0.00005
R4 = {120*(1+strain)}
```

The simulation confirmed a linear differential output ranging from $-2.5\,\mathrm{mV}$ to $+2.5\,\mathrm{mV}$ across the swept strain range, consistent with theoretical predictions.

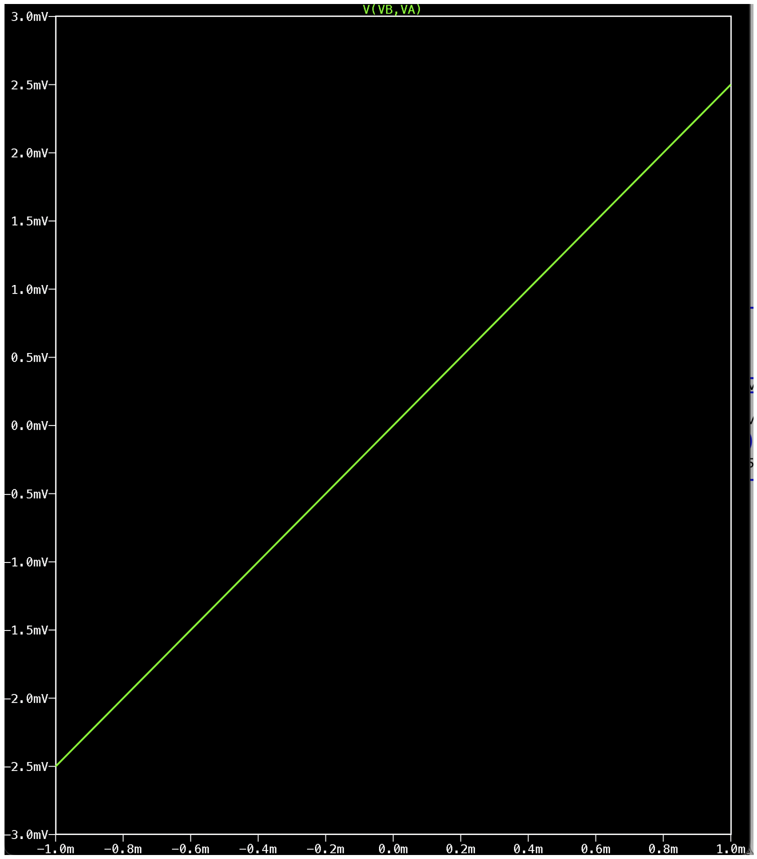

Figure 3: LTspice DC sweep result showing the differential bridge output $V(A) - V(B)$ as a function of the strain parameter. The linear response from $-2.5\,\mathrm{mV}$ to $+2.5\,\mathrm{mV}$ confirms correct bridge operation.

3.4 Bridge Balancing Challenges

In practice, the fixed resistors measured $98\,\Omega$ and $99\,\Omega$ respectively (rather than an ideal $100\,\Omega$), and the two strain gauges had slightly different nominal resistances. This resulted in a non-zero differential output at rest, observed as approximately $-12\,\mathrm{mV}$ steady-state offset. This was addressed in software using the HX711 tare function, which captures and subtracts the resting output from all subsequent readings. To stick the strain gauge resistors onto the metal sheet was a tedious process. We had to sandpaper the metal sheet and make it as smooth as possible to ensure there were no bubbles after sticking it.

4 Custom Instrumentation Amplifier

4.1 Motivation and Design Choice

An instrumentation amplifier (INA) is required to amplify the small differential bridge output while rejecting common-mode interference. A discrete three-op-amp INA topology was implemented in hardware due to its high common-mode rejection ratio (CMRR), differential gain control via a single resistor, and high input impedance which minimises bridge loading.

The AD620 was used as the reference model during LTspice simulation, with the hardware implementation realised using discrete op-amps in the classical three-op-amp INA configuration.

4.2 Circuit Topology

The three-op-amp INA consists of two input buffer amplifiers (U1, U2) followed by a differential output stage (U3), as shown in Fig. 10.

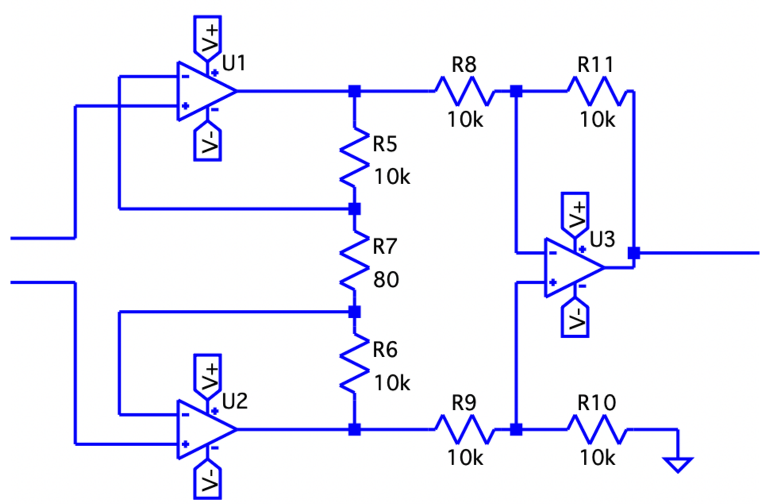

Figure 4: Custom three-op-amp instrumentation amplifier implemented in LTspice. U1 and U2 act as input buffers, while U3 forms the differential output stage. Gain is controlled by R_G.

4.3 Gain Calculation

The gain of the three-op-amp INA is given by:

$$G = 1 + \frac{2R_f}{R_G} \tag{5}$$

where R_f is the feedback resistor and R_G is the gain-setting resistor. In the hardware implementation, R_G was realised as a $40\,\Omega$ fixed resistor in series with a $5\,\mathrm{k}\Omega$ potentiometer, enabling continuously variable gain.

The required gain was derived from the desired output voltage range. To swing the output over a useful range of $\approx 3\,\mathrm{V}$ from a $4.7\,\mathrm{mV}$ bridge signal:

$$G_{required} = \frac{V_{out}}{V_{diff}} = \frac{3}{0.0047} \approx 638 \tag{6}$$

In practice, the gain was tuned empirically via the potentiometer (it increases or decreases the value of R_G) to avoid saturation while maximising dynamic range.

4.4 Common-Mode Rejection

The INA topology inherently rejects common-mode signals. With matched resistors, the CMRR is theoretically infinite; in practice, resistor mismatch limits CMRR to typically 80 dB for discrete implementations. This is sufficient to reject 50 Hz mains interference and switching noise from the ESP32's WiFi radio, both of which appear as common-mode signals on the bridge output lines.

4.5 Simulation Results

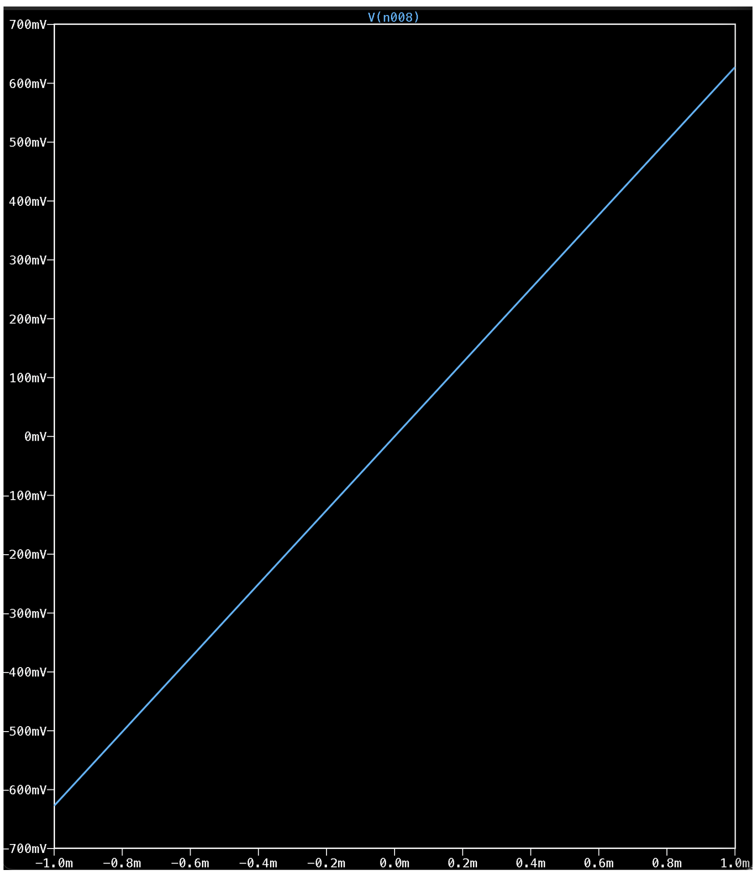

Figure 5: LTspice simulation of the instrumentation amplifier output versus strain sweep. The amplified output exhibits a linear response consistent with the designed gain and is centered around 0V.

5 Signal Conditioning: Inverting Summer and Low-Pass Filter

5.1 Overview

Following amplification, the signal requires two further processing steps before digitisation:

1. **DC offset correction:** The INA output carries a DC offset due to bridge imbalance and the REF pin voltage. An inverting summing amplifier is used to subtract this offset, centering the signal around zero.

2. **Low-pass filtering:** High-frequency noise from power supply ripple, mechanical vibration, and electromagnetic interference must be attenuated. An active low-pass filter with gain achieves this.

5.2 Inverting Summing Amplifier

The inverting summing amplifier combines the INA output with a precision DC reference voltage ($V_{ref} = 0.6\,\mathrm{V}$), generated by a unity-gain voltage buffer (U6) driven by a resistor divider from the 5 V supply:

$$V_{ref} = V_{supply} \times \frac{R_{15}}{R_{14} + R_{15}} = 5 \times \frac{12.5\,\mathrm{k}}{87.5\,\mathrm{k} + 12.5\,\mathrm{k}} = 0.625\,\mathrm{V} \tag{7}$$

The summing amplifier output is:

$$V_{sum} = -\left(\frac{R_f}{R_{in1}} V_{INA} + \frac{R_f}{R_{in2}} V_{ref}\right) \tag{8}$$

With $R_{in1} = R_{in2} = R_f = 47\,\mathrm{k}\Omega$, this gives unity summing gain with sign inversion, effectively subtracting the reference from the INA output.

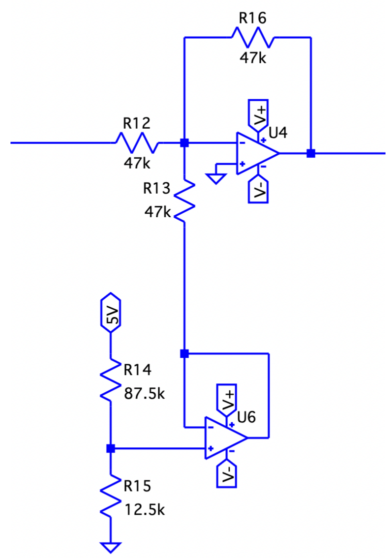

Figure 6: LTspice schematic of the inverting summing amplifier and voltage reference buffer. R12, R13, R16 = 47 kΩ; R14 = 87.5 kΩ; R15 = 12.5 kΩ.

5.3 Active Low-Pass Filter with Gain

An active Sallen-Key low-pass filter with additional gain was implemented following the summing stage. The cutoff frequency is given by:

$$f_c = \frac{1}{2\pi RC} \tag{9}$$

For $R = 10\,\mathrm{k\Omega}$ and $C = 1\,\mu\mathrm{F}$:

$$f_c = \frac{1}{2\pi \times 10\,\mathrm{k} \times 1\,\mu} \approx 15.9\,\mathrm{Hz} \tag{10}$$

This cutoff frequency is appropriate for quasi-static load measurement: structural loads change slowly (well below 10 Hz), while the dominant noise sources (50 Hz mains, WiFi switching at MHz frequencies) are well above the passband. The filter attenuates these at −20 dB/decade.

An additional gain stage ($R_{17} = 24\,\mathrm{k\Omega}$, $R_{18} = 10\,\mathrm{k\Omega}$, $C_2 = 660\,\mathrm{nF}$) follows the filter to scale the output to a 0–3 V range suitable for the subsequent ADC stage.

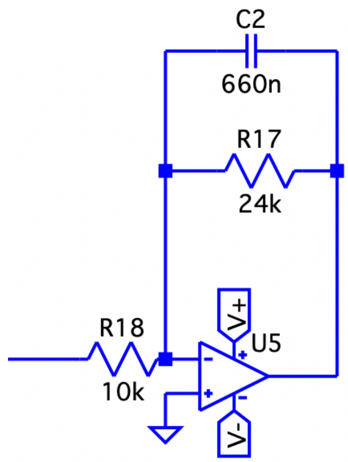

Figure 7: LTspice schematic of the active low-pass filter with gain stage. R17 = 24 kΩ, R18 = 10 kΩ, C2 = 660 nF.

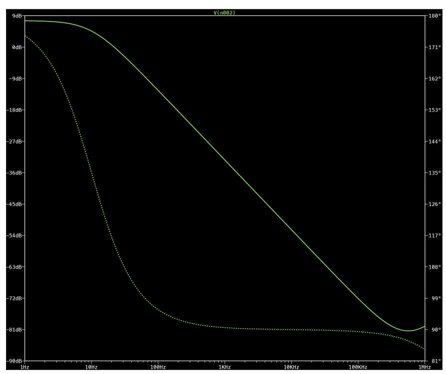

Figure 8: AC analysis of the low-pass filter stage, showing the −3 dB cutoff at approximately 15.9 Hz and −20 dB/decade roll-off beyond the passband.

5.4 Output Range Verification

The complete analog signal chain produces an output in the range of 0 V to 3 V, verified in both simulation and hardware. At zero strain, the output is centred near 1.5 V; at maximum compressive strain the output approaches 0 V, and at maximum tensile strain it approaches 3 V.

6 IoT Implementation

6.1 System Architecture

The IoT subsystem was designed to transmit weight measurements wirelessly to a live browser dashboard with minimal latency. The architecture is entirely local — no cloud services are used — ensuring low latency, privacy, and operation without internet connectivity. The stack comprises:

- **HX711:** 24-bit sigma-delta ADC with integrated programmable gain amplifier, designed specifically for Wheatstone bridge interfacing.
- **ESP32:** Dual-core 240 MHz microcontroller with integrated WiFi, running the Arduino framework.
- **Mosquitto:** Open-source MQTT broker running locally on a laptop.
- **Node-RED:** Flow-based visual programming environment hosting the dashboard at `http://localhost:1880/ui`.

6.2 HX711 ADC Interface

The HX711 was selected over the ESP32's internal 12-bit SAR ADC for two reasons: its 24-bit resolution provides significantly finer quantisation (16 million counts versus 4096), and its dedicated differential input is optimised for Wheatstone bridge signals, with an internal PGA supporting gains of 32, 64, and 128.

In this implementation, the bridge output was connected directly to the HX711 Channel A inputs (A+, A-) at a gain of 128, bypassing the external instrumentation amplifier for the final hardware configuration. The HX711 effective input range at gain 128 and $V_{supply} = 4.5\,\mathrm{V}$ is:

$$V_{in,max} = \pm\frac{V_{supply}}{2 \times Gain} = \pm\frac{4.5}{2 \times 128} \approx \pm 17.6\,\mathrm{mV} \tag{11}$$

The HX711 communicates with the ESP32 via a custom two-wire serial protocol using a clock pin (SCK, GPIO5) and data pin (DT, GPIO4).

6.3 MQTT Communication

Weight data is published to the MQTT topic `strain/weight` at 2 Hz. The ESP32 acts as an MQTT client connecting to the Mosquitto broker at the laptop's local IP address over the shared WiFi hotspot. Node-RED subscribes to this topic and updates the dashboard gauge widget in real time.

The MQTT protocol was selected for its low overhead, publish-subscribe architecture, and native support in both the ESP32 Arduino ecosystem (PubSubClient library) and Node-RED.

6.4 Calibration Procedure

Empirical calibration was performed by placing objects of known weight on the beam and recording the corresponding raw HX711 counts after taring. The scale factor k (counts per gram) is defined as:

$$k = \frac{C_{raw}}{m} \tag{12}$$

where C_{raw} is the tared raw count and m is the known mass in grams. The calibration data is presented in Table 1.

Table 1: Empirical Calibration Data

Object	Mass (g)	Raw Count	Scale Factor (counts/g)
iPhone 16 Pro	210	7000	33.3
JBL Go 2	184	6750	36.7
AirPods Pro 3	55	5509	100.2
Water Bottle (300 ml)	310	7961	25.7
Average	–	–	**49.0**

The variance in scale factor across the calibration objects reflects several practical factors: the nonlinearity of the beam's mechanical response at different load levels, the sensitivity of the half-bridge to load placement position on the beam, and the thermal drift of the gauge resistance between measurements. A more robust calibration would involve a polynomial fit across a wider and denser set of reference masses, which is identified as future work.

6.5 Firmware

The ESP32 firmware was developed in the Arduino framework. The core measurement loop reads averaged HX711 counts, applies the scale factor, publishes the result to MQTT, and prints to the serial monitor simultaneously. A software tare routine captures the resting baseline count on demand and subtracts it from all subsequent readings, compensating for bridge imbalance and thermal offset.

Listing 1: Core ESP32 firmware loop

```
void loop() {
  if (!client.connected()) reconnect();
  client.loop();

  if (scale.is_ready()) {
    long raw = scale.read_average(20);
    long adjusted = raw - zero_offset;
    float weight = adjusted / calibration_factor;
    char msg[50];
    dtostrf(weight, 6, 2, msg);
```

```
      client.publish("strain/weight", msg);
      Serial.print("Weight: ");
      Serial.print(weight);
      Serial.println(" g");
    }
    delay(500);
  }
```

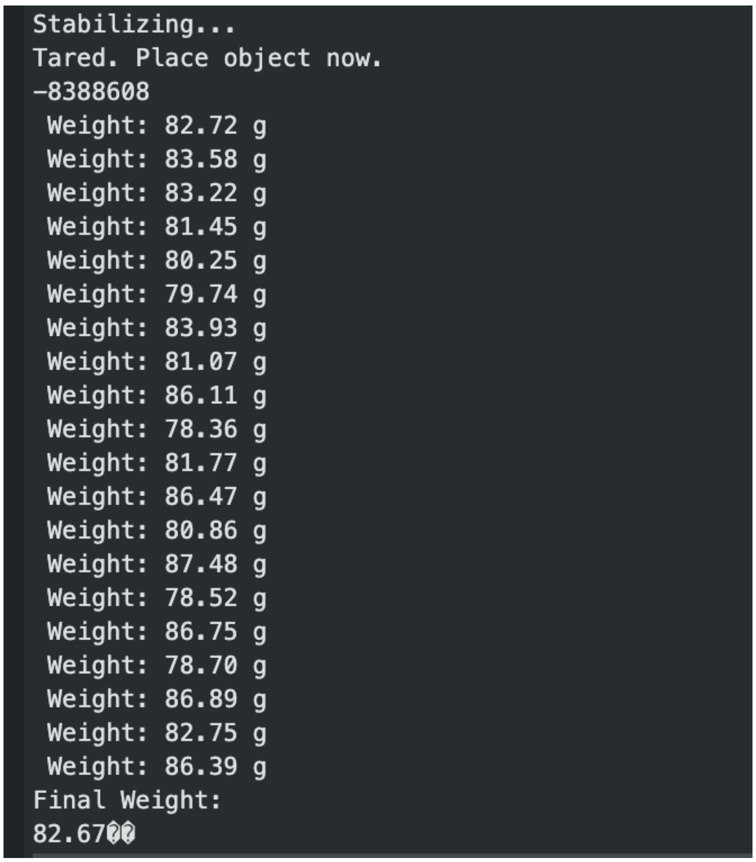

Figure 9: Real-time weight of the adapter measured by the strain gauge on serial monitor

7 Challenges and Solutions

The development process surfaced several non-trivial engineering challenges, each of which required systematic diagnosis and resolution.

7.1 Thermal Drift

Observation: On power-up, HX711 raw counts drifted from approximately +50,000 to near zero over a 15–20 minute period, even without any applied load.

Root Cause: Self-heating of the strain gauges due to excitation current. The excitation current through a 120 Ω gauge at 4.5 V generates ohmic heating, raising the gauge temperature and consequently its resistance. This is a well-known phenomenon in strain gauge metrology called self-heating error.

Solution: A mandatory warm-up period of 15–20 minutes was incorporated into the measurement procedure. The software tare is triggered only after readings stabilise, rather than automatically on startup.

7.2 Ground Loop Noise

Observation: Random, high-amplitude fluctuations (± 1000 counts) in HX711 readings that did not correlate with beam deflection.

Root Cause: Two separate ground wires were routed from the ESP32 to the HX711 (one to the GND pin, one to E-). This created a ground loop: small current differences between the two return paths generated a fluctuating common-mode voltage that was picked up as a differential signal.

Solution: Both HX711 GND and E- were connected to a single ground node, with only one wire returning to the ESP32 GND pin, eliminating the loop.

7.3 Bridge Imbalance

Observation: A steady-state differential output of approximately $-12\,\mathrm{mV}$ was observed even at zero applied load.

Root Cause: The two fixed bridge resistors measured $98\,\Omega$ and $99\,\Omega$ (rather than ideal matching), and the two strain gauges had different nominal resistances, resulting in an inherently unbalanced bridge.

Solution: The offset was compensated in software using the HX711 tare function, which captures and subtracts the resting raw count from all subsequent measurements.

7.4 LTspice Symbol Pin Mapping Error

Observation: The simulated AD620 instrumentation amplifier produced a gain of less than unity (attenuation) rather than the designed gain of ≈ 144.

Root Cause: When a custom symbol was drawn for the AD620 in LTspice, the `SpiceOrder` attribute of each pin in the `.asy` file did not correctly match the argument order of the `.SUBCKT` definition. Specifically, the R_G pin was mapped to the IN+ subcircuit node, effectively shorting the input and collapsing the gain.

Solution: The `.asy` file was corrected by explicitly assigning `SpiceOrder` values to each pin to match the `.SUBCKT AD620 1 2 99 50 46 20 7 8` argument order precisely.

7.5 HX711 Input Saturation Risk

Observation: The initial design routed the AD620 output (amplified to several volts) directly into the HX711 analog input.

Root Cause: Misunderstanding of the HX711 input architecture. The HX711 Channel A differential input range at gain 128 is only $\pm 17.6\,\mathrm{mV}$. Feeding an amplified signal of several volts would have saturated or damaged the input stage.

Solution: The external INA was bypassed for the hardware implementation, and the bridge was connected directly to the HX711 inputs, utilising the HX711's internal PGA for amplification.

8 Experimental Results

8.1 Resistance Change Characterisation

The strain gauge resistance was measured using a precision multimeter while progressively loading the cantilever beam. The nominal resistance at rest was $121.8\,\Omega$. At maximum applied deflection (limited by beam geometry), a resistance change of $\pm 0.5\,\Omega$ was observed on each gauge, corresponding to a strain of $\varepsilon \approx 2050\,\mu\varepsilon$ as calculated in (2).

8.2 HX711 Raw Count Response

After taring, the HX711 raw counts increased monotonically with applied load, confirming the correct polarity and monotonicity of the measurement chain. Representative steady-state readings at rest and under load were stable within ± 15 counts after the warm-up period, corresponding to a weight uncertainty of approximately $\pm 0.4\,$g at the nominal scale factor.

8.3 Calibration Results

The calibration data in Table 1 reveals a spread in scale factor from 25.7 to 100.2 counts/-gram. This spread is attributed to:

- **Load position sensitivity:** The beam's mechanical response is position-dependent. A load placed closer to the fixed end produces less deflection and hence fewer counts per gram than the same load at the free end.
- **Thermal drift between measurements:** The baseline count drifted between successive calibration measurements, introducing systematic error in the tared readings.
- **Nonlinearity at low loads:** At very low loads (e.g., AirPods Pro at 55 g), the signal-to-noise ratio is lower, and the scale factor estimate is less reliable.

8.4 Dashboard Visualisation

The Node-RED dashboard displayed live weight readings on a gauge widget, updating at 2 Hz over MQTT. The dashboard was successfully accessed from a second device connected to the same WiFi hotspot, demonstrating the networked IoT capability of the system.

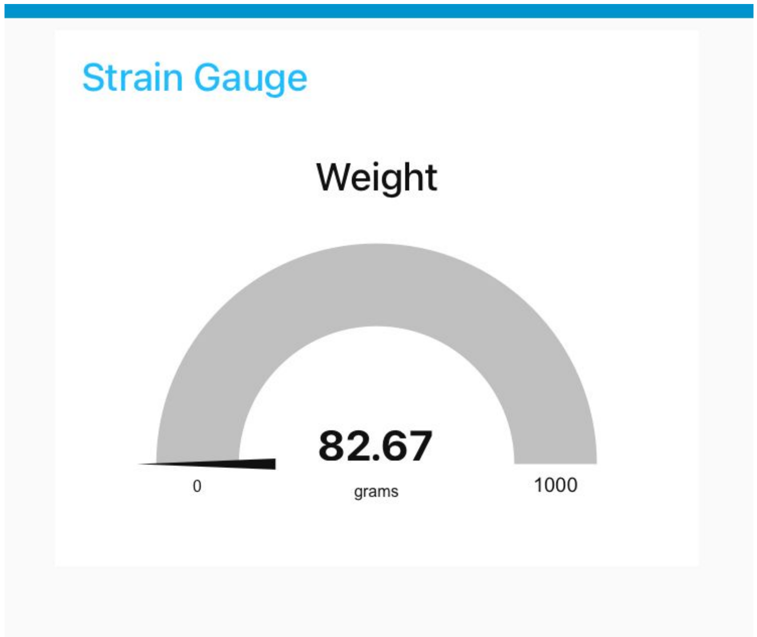

Figure 10: Weight of the Adapter

9 Discussion/Conclusion

These are the links for the videos that test the weight of the adapter and that shows the working of the wheatstone bridge converting the resistance change to voltage change. Watch Multimeter Readings.

Watch Serial Monitor Readings.

The implemented system successfully demonstrates end-to-end strain-based weight measurement with IoT integration. However, several limitations of the current implementation merit discussion.

The most significant practical challenge was thermal drift, which required a lengthy warm-up period and precluded rapid successive measurements. In a production system, this would be addressed through the use of a dummy gauge (a gauge bonded transversely to the beam, unresponsive to longitudinal strain but equally affected by temperature) as the fourth arm of a full bridge, providing continuous passive temperature compensation.

The single-point empirical calibration (using a fixed scale factor) is adequate for demonstration but introduces systematic error across the load range. A polynomial regression over a larger set of calibration points would yield significantly better accuracy, particularly at the extremes of the measurement range.

The use of a breadboard for prototyping introduced intermittent contact failures, particularly at the high-impedance HX711 differential inputs. Migration to a soldered perfboard or PCB would substantially improve reliability and reduce susceptibility to mechanical and thermal disturbance.